Lecture 3: Stationary Time-Series Processes II

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Announcement: Assignments

- The assignment schedule has been updated!

	Topics	Posted	Due Date
Assignment 1	1.	27/02	20/03
Assignment 2	2., 3., and 4.	27/03	20/04
Assignment 3	5., 6., and 7.	30/04	22/05

- Assignment 1 will be uploaded by the end of today.
  - Goal 1: Review what we have studied so far.
  - Goal 2: Study the consequences of trend and seasonal components.

Recap

Last Class

Time Series

- Stationarities:
  - Strong stationarity, weak stationarity.
- White noise processes:
  - WN, IWN, GWN.
- Random walk:

$$X_t = X_{t-1} + \varepsilon_t \qquad \Longleftrightarrow \qquad X_t = X_0 + \sum_{s=1}^{t} \varepsilon_s,$$

where $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$.

Outline

- Preliminary: Math Essentials
- Moving Average (MA) processes
- Autoregressive (AR) processes
  - characteristic function, MA representation
- Autoregressive Moving Average (ARMA) processes
- Gateway to Econometric Analysis

Stationary Time-Series Processes

Math Essentials

Math Essentials

- Finite geometric sum of a constant term a :

$$\sum_{j=0}^{J-1} a^j = 1 + a + a^2 + a^3 + \cdots + a^{J-1} = \begin{cases} J & \text{for } a = 1 \\ \frac{1-a^J}{1-a} & \text{for } a \neq 1. \end{cases}$$

- Infinite geometric decreasing sum of a constant term a:

$$\sum_{j=0}^{\infty} a^j = 1 + a + a^2 + a^3 + a^4 + \cdots = \frac{1}{1-a}, \quad \text{for } |a| < 1.$$

Stationary Time-Series Processes

Moving Average (MA) Process

MA(q) Process

Definition (Moving Average of Order q)

A time series $\{X_t\}_{t=-\infty}^{\infty}$ is said to follow an MA(q) process if

$$X_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-q},$$

where $\mu, \theta_1, \ldots, \theta_{t-q} \in \mathbb{R}$ are constants, and $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$.

- An MA(q) process is not affected by shocks from far past beyond q lags.
  - a short memory process
- θ_s measures the persistence of the s-lagged shock.
- $\mu, \theta_1, \ldots, \theta_{t-q}, \sigma_\varepsilon^2$: parameters

MA(1) Process: Mean

- Given $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$, $\{X_t\}_{t=-\infty}^{\infty}$ is an MA(1) process if

$$X_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}.$$

- Q. Is an MA(1) process strongly stationary or weakly stationary?

- **Mean:**

$$\begin{aligned} E[X_t] &= E[\mu + \varepsilon_t + \theta\varepsilon_{t-1}] \\ &= \mu + \underbrace{E[\varepsilon_t]}_{0} + \theta \underbrace{E[\varepsilon_{t-1}]}_{0} \\ &= \mu. \end{aligned}$$

MA(1) Process: Variance

- Given $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$, $\{X_t\}_{t=-\infty}^{\infty}$ is an MA(1) process if

$$X_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}.$$

- **Variance:**

$$\begin{aligned} Var(X_t) &= E[(X_t - \mu)^2] = E[(\varepsilon_t + \theta\varepsilon_{t-1})^2] \\ &= E[\varepsilon_t^2 + \theta^2\varepsilon_{t-1}^2 + 2\theta\varepsilon_t\varepsilon_{t-1}] \\ &= \underbrace{E[\varepsilon_t^2]}_{\sigma_\varepsilon^2} + \theta^2 \underbrace{E[\varepsilon_{t-1}^2]}_{\sigma_\varepsilon^2} + 2\theta \underbrace{E[\varepsilon_t\varepsilon_{t-1}]}_{0} \\ &= (1+\theta^2)\sigma_\varepsilon^2. \end{aligned}$$

MA(1) Process: Autocovariance

- Given $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$, $\{X_t\}_{t=-\infty}^{\infty}$ is an MA(1) process if

$$X_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1},$$

- **Autocovariance:** for lag $h = 1$,

$$\begin{aligned}
Cov(X_t, X_{t-1}) &= E[(X_t - \mu)(X_{t-1} - \mu)] \\
&= E[(\varepsilon_t + \theta\varepsilon_{t-1})(\varepsilon_{t-1} + \theta\varepsilon_{t-2})] \\
&= E[\varepsilon_t\varepsilon_{t-1} + \theta\varepsilon_t\varepsilon_{t-2} + \theta\varepsilon_{t-1}^2 + \theta^2\varepsilon_{t-1}\varepsilon_{t-2}] \\
&= \underbrace{E[\varepsilon_t\varepsilon_{t-1}]}_{0} + \theta\underbrace{E[\varepsilon_t\varepsilon_{t-2}]}_{0} + \theta\underbrace{E[\varepsilon_{t-1}^2]}_{\sigma_\varepsilon^2} + \theta^2\underbrace{E[\varepsilon_{t-1}\varepsilon_{t-2}]}_{0} \\
&= \theta\sigma_\varepsilon^2.
\end{aligned}$$

MA(1) Process: Autocovariance

- Given $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$, $\{X_t\}_{t=-\infty}^{\infty}$ is an MA(1) process if

$$X_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1},$$

- **Autocovariance:** for lag $h = 2$,

$$\begin{aligned}
Cov(X_t, X_{t-2}) &= E[(X_t - \mu)(X_{t-2} - \mu)] \\
&= E[(\varepsilon_t + \theta\varepsilon_{t-1})(\varepsilon_{t-2} + \theta\varepsilon_{t-3})] \\
&= E[\varepsilon_t\varepsilon_{t-2} + \theta\varepsilon_t\varepsilon_{t-3} + \theta\varepsilon_{t-1}\varepsilon_{t-2} + \theta^2\varepsilon_{t-1}\varepsilon_{t-3}] \\
&= \underbrace{E[\varepsilon_t\varepsilon_{t-2}]}_{0} + \theta \underbrace{E[\varepsilon_t\varepsilon_{t-3}]}_{0} + \theta \underbrace{E[\varepsilon_{t-1}\varepsilon_{t-2}]}_{0} + \theta^2 \underbrace{E[\varepsilon_{t-1}\varepsilon_{t-3}]}_{0} \\
&= 0.
\end{aligned}$$

- Likewise, for any lag $h \geq 3$,

$$Cov(X_t, X_{t-h}) = 0.$$

MA(1) Process: Autocorrelation

- Given $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$, $\{X_t\}_{t=-\infty}^{\infty}$ is an MA(1) process if

$$X_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1},$$

- **Autocorrelation:**
  - $\rho_{X,h}$ for $h = 1$:

$$\begin{aligned}
\rho_{X,1} := Corr(X_t, X_{t-1}) &= \frac{Cov(X_t, X_{t-1})}{\sqrt{Var(X_t)}\sqrt{Var(X_{t-1})}} \\
&= \frac{\theta\sigma_\varepsilon^2}{\sqrt{(1+\theta^2)\sigma_\varepsilon^2}\sqrt{(1+\theta^2)\sigma_\varepsilon^2}} = \frac{\theta\sigma_\varepsilon^2}{(1+\theta^2)\sigma_\varepsilon^2} \\
&= \frac{\theta}{1+\theta^2}
\end{aligned}$$

MA(1) Process: Autocorrelation

- Given $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$, $\{X_t\}_{t=-\infty}^{\infty}$ is an MA(1) process if

$$X_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1},$$

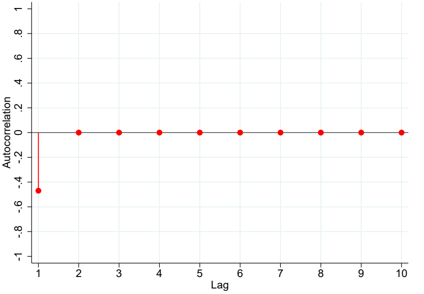
- **Autocorrelation:**
  - $\rho_{X,h}$ for $h \geq 2$:

$$\begin{aligned}
\rho_{X,h} &:= Corr(X_t, X_{t-h}) = \frac{Cov(X_t, X_{t-h})}{\sqrt{Var(X_t)}\sqrt{Var(X_{t-h})}} \\
&= \frac{0}{\sqrt{(1+\theta^2)\sigma_\varepsilon^2}\sqrt{(1+\theta^2)\sigma_\varepsilon^2}} = \frac{0}{(1+\theta^2)\sigma_\varepsilon^2} \\
&= 0
\end{aligned}$$

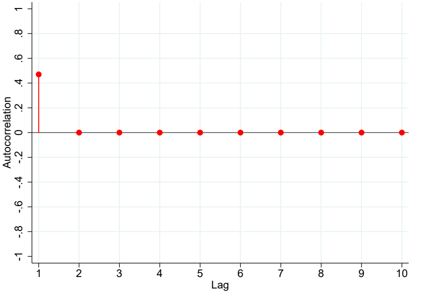
- Autocorrelation $\rho_{X,h}$ depends on the lag h.
  - Autocorrelation function (ACF): a plot of $\rho_{X,h}$ with respect to lag h.
  - a.k.a. Correlogram.

MA(1) Process: Autocorrelation

- **ACF** for MA(1)



(a) $\theta = 0.7$



(b) $\theta = -0.7$

MA(1) Process: Summary

- Given $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$, $\{X_t\}_{t=-\infty}^{\infty}$ is an MA(1) process if

$$X_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1}.$$

  Mean: $E[X_t] = \mu$

  Variacne: $Var(X_t) = (1 + \theta^2)\sigma_\varepsilon^2$

  Autocovariance: $Cov(X_t, X_{t-h}) = \begin{cases} \theta\sigma_\varepsilon^2 & \text{for } h = 1 \\ 0 & \text{for } h \geq 2 \end{cases}$

- This process has a short memory (only one-period memory).
- An MA(1) process is **weakly stationary**.
- An MA(1) process is not **strongly stationary**.
  - It is **strongly stationary** if $\varepsilon_s \sim IWN(0, \sigma_\varepsilon^2)$ for all $s \leq t$. (Why?)

Stationary Time-Series Processes

Autoregressive (AR) Process

AR(p) Process

Definition (Auto Regressive of Order p)

A time series $\{X_t\}_{t=-\infty}^{\infty}$ is said to be an AR(p) process if

$$X_t = \alpha + \rho_1 X_{t-1} + \rho_2 X_{t-2} + \ldots + \rho_p X_{t-p} + \varepsilon_t,$$

where $\alpha, \rho_1, \ldots, \rho_p \in \mathbb{R}$ are constants, and $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$.

- A shock ε_t can exert long-lasting effects through lagged dependent variables.
  - a long-memory process
- ρ_s measures the persistence of the s-lagged variable.
- $\alpha, \rho_1, \ldots, \rho_p, \sigma_\varepsilon^2$: parameters

AR(1) Process

- Given $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$, $\{X_t\}_{t=-\infty}^{\infty}$ is an **AR(1)** process if

$$X_t = \alpha + \rho X_{t-1} + \varepsilon_t.$$

- Suppose $|\rho| < 1$.

  $\Longrightarrow$ An AR(1) process is weakly stationary.

- Q. How do we know this?

  A. Characteristic function!

AR(1) Process: Characteristic Function

- An **AR(1)** process:

$$X_t = \alpha + \rho X_{t-1} + \varepsilon_t.$$

- The characteristic equation:

$$1 - \rho z = 0,$$

  where $z \in \mathbb{C}$.

- Let z^* be the root of this equation.

- **Proposition:** An AR(1) process is weakly stationary if and only if $|z^*| > 1$.

AR(1) Process: Characteristic Function

- **Proposition:** An AR(1) process is weakly stationary if and only if $|z^*| > 1$.

- **Application:**

  An AR(1) process is weakly stationary

  $\iff 1 - \rho z^* = 0$ with $|z^*| > 1$

  $\iff z^* = \frac{1}{\rho}$ with $|z^*| > 1$

  $\iff \rho = \frac{1}{z^*}$ with $|z^*| > 1$ (i.e., $|\rho| < 1$)

AR(1) Process: Characteristic Function

- An AR(p) process,

$$X_t = \alpha + \rho_1 X_{t-1} + \ldots + \rho_p X_{t-p} + \varepsilon_t.$$

- The corresponding characteristic function is given by

$$1 - \rho_1 z^1 - \ldots - \rho_p z^p = 0.$$

- Technique:

$$\underbrace{X_{t-0}}_{z^0} - \rho_1 \underbrace{X_{t-1}}_{z^1} - \ldots - \rho_p \underbrace{X_{t-p}}_{z^p} = \underbrace{\alpha + \varepsilon_t}_{0}$$
$$\therefore 1 - \rho_1 z - \ldots - \rho_p z^p = 0$$

- An AR(p) process is weakly stationary.
$\iff$ All the roots of the characteristic function are larger in magnitude than one.

AR(1) Process: Characteristic Function

- Ex) An AR(2) process:

$$X_t = \rho_0 + \rho_1 X_{t-1} + \rho_2 X_{t-2} + \varepsilon_t,$$

where $\rho_1 = 7\rho$ and $\rho_2 = 12\rho^2$ with $\rho > 0$, and $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$.

- The associated characteristic function is

$$1 - 7\rho z - 12\rho^2 z^2 = 0$$
$$\iff (1 - 3\rho z)(1 - 4\rho z) = 0.$$

- The roots z^* are $\frac{1}{3\rho}$ and $\frac{1}{4\rho}$.
- This process is weakly stationary.
$\iff |\frac{1}{3\rho}| > 1 \quad \wedge \quad |\frac{1}{4\rho}| > 1$
$\iff \frac{1}{3\rho} > 1 \quad \wedge \quad \frac{1}{4\rho} > 1$
$\iff \rho < \frac{1}{3} \quad \wedge \quad \rho < \frac{1}{4}$
- Hence, the process is weakly stationary if and only if $(0 <)\rho < \frac{1}{4}$.

MA(∞) Representation of AR(1) Process

- A stationary **AR(1)** (i.e., $|\rho| < 1$) $\{X_t\}_{t=-\infty}^{\infty}$ can be represented as an **MA(∞)**:

$$X_t = \frac{\alpha}{1-\rho} + \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j}$$

Proof: By repeated substitution,

$$\begin{aligned}
X_t &= \alpha + \rho \underbrace{X_{t-1}}_{\alpha+\rho X_{t-2}+\varepsilon_{t-1}} + \varepsilon_t \\
&= \alpha + \rho(\alpha + \rho X_{t-2} + \varepsilon_{t-1}) + \varepsilon_t \\
&= \alpha + \alpha\rho + \rho^2 \underbrace{X_{t-2}}_{\alpha+\rho X_{t-3}+\varepsilon_{t-2}} + \rho\varepsilon_{t-1} + \varepsilon_t \\
&= \alpha + \alpha\rho + \rho^2(\alpha + \rho X_{t-3} + \varepsilon_{t-2}) + \rho\varepsilon_{t-1} + \varepsilon_t \\
&= \alpha + \alpha\rho + \alpha\rho^2 + \rho^3 X_{t-3} + \rho^2\varepsilon_{t-2} + \rho\varepsilon_{t-1} + \varepsilon_t \\
&= \dots
\end{aligned}$$

MA(∞) Representation of AR(1) Process

Proof (cont'd):

$$
\begin{aligned}
X_t &= \alpha(1 + \rho + \rho^2 + \ldots) + \varepsilon_t + \rho\varepsilon_{t-1} + \rho^2\varepsilon_{t-2} + \ldots \\
&= \alpha \underbrace{\sum_{j=0}^{\infty} \rho^j}_{\frac{1}{1-\rho}} + \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j} \\
&= \frac{\alpha}{1-\rho} + \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j}.
\end{aligned}
$$

$\square$

- In general, **any** stationary **AR** process can be expressed as an **MA(∞)** process.
  - The Wold decomposition
- The opposite is also true: Any MA(q) process is "invertible" to an AR(∞) process.

AR(1) Process: Mean

- The **MA(∞)** representation of an **AR(1)** process (assuming $|\rho| < 1$):

$$X_t = \frac{\alpha}{1-\rho} + \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j},$$

where $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$.

- **Mean:**

$$\begin{aligned} E[X_t] &= E\left[\frac{\alpha}{1-\rho} + \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j}\right] \\ &= \frac{\alpha}{1-\rho} + \sum_{j=0}^{\infty} \rho^j \underbrace{E[\varepsilon_{t-j}]}_{0} \\ &= \frac{\alpha}{1-\rho}. \end{aligned}$$

AR(1) Process: Variance

- The **MA(∞)** representation of an **AR(1)** process (assuming $|\rho| < 1$):

$$X_t = \frac{\alpha}{1-\rho} + \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j},$$

where $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$.

- **Variance:**

$$\begin{aligned}
Var(X_t) = E\left[\left(X_t - \frac{\alpha}{1-\rho}\right)^2\right] &= E\left[\left(\sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j}\right)^2\right] \\
&= E[\varepsilon_t^2 + \rho^2\varepsilon_{t-1}^2 + \rho^4\varepsilon_{t-2}^2 + \rho^6\varepsilon_{t-3}^2 + \ldots \\
&\quad + 2(\rho\varepsilon_t\varepsilon_{t-1} + \rho^2\varepsilon_t\varepsilon_{t-2} + \rho^3\varepsilon_t\varepsilon_{t-3} + \ldots)] \\
&= E[\varepsilon_t^2] + \rho^2 E[\varepsilon_{t-1}^2] + \rho^4 E[\varepsilon_{t-2}^2] + \rho^6 E[\varepsilon_{t-3}^2] + \ldots \\
&\quad + 2\Big(\rho E[\varepsilon_t\varepsilon_{t-1}] + \rho^2 E[\varepsilon_t\varepsilon_{t-2}] + \rho^3 E[\varepsilon_t\varepsilon_{t-3}] + \ldots\Big)
\end{aligned}$$

AR(1) Process: Variance

- **Variance (cont'd):**

$$
\begin{aligned}
Var(X_t) &= \underbrace{Var(\varepsilon_t)}_{\sigma_\varepsilon^2} + \rho^2 \underbrace{Var(\varepsilon_{t-1})}_{\sigma_\varepsilon^2} + \rho^4 \underbrace{Var(\varepsilon_{t-2})}_{\sigma_\varepsilon^2} + \rho^6 \underbrace{Var(\varepsilon_{t-3})}_{\sigma_\varepsilon^2} + \dots \\
&\qquad + 2\Big(\rho \underbrace{Cov(\varepsilon_t, \varepsilon_{t-1})}_{0} + \rho^2 \underbrace{Cov(\varepsilon_t, \varepsilon_{t-2})}_{0} + \rho^3 \underbrace{Cov(\varepsilon_t, \varepsilon_{t-3})}_{0} + \dots \Big) \\
&= \sigma_\varepsilon^2 (1 + \rho^2 + \rho^4 + \rho^6 + \dots) \\
&= \sigma_\varepsilon^2 \underbrace{\sum_{j=0}^{\infty} (\rho^2)^j}_{\frac{1}{1-\rho^2}} \\
&= \frac{\sigma_\varepsilon^2}{1-\rho^2}.
\end{aligned}
$$

AR(1) Process: Autocorrelation function

- The **MA(∞)** representation of an **AR(1)** process (assuming $|\rho| < 1$):

$$X_t = \frac{\alpha}{1-\rho} + \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j},$$

where $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$.

- **Autocovariance ($h = 1$):**

$$\begin{aligned}
Cov(X_t, X_{t-1}) &= E\Big[\Big(X_t - \frac{\alpha}{1-\rho}\Big)\Big(X_{t-1} - \frac{\alpha}{1-\rho}\Big)\Big] \\
&= E\Big[\Big(\sum_{i=0}^{\infty} \rho^i \varepsilon_{t-i}\Big)\Big(\sum_{j=0}^{\infty} \rho^j \varepsilon_{(t-1)-j}\Big)\Big] = E\Big[\Big(\varepsilon_t + \rho \sum_{j=0}^{\infty} \rho^j \varepsilon_{(t-1)-j}\Big)\Big(\sum_{j=0}^{\infty} \rho^j \varepsilon_{(t-1)-j}\Big) \\
&= E\Big[\sum_{j=0}^{\infty} \rho^j \varepsilon_t \varepsilon_{(t-1)-j} + \rho \Big(\sum_{j=0}^{\infty} \rho^j \varepsilon_{(t-1)-j}\Big)^2\Big]
\end{aligned}$$

AR(1) Process: Autocorrelation function

- **Autocovariance ($h = 1$, cont'd):**

$$
\begin{aligned}
Cov(X_t, X_{t-1}) &= \sum_{j=0}^{\infty} \rho^j E[\varepsilon_t \varepsilon_{(t-1)-j}] + \rho E\left[\left(\sum_{j=0}^{\infty} \rho^j \varepsilon_{(t-1)-j}\right)^2\right] \\
&= \sum_{j=0}^{\infty} \rho^j \underbrace{Cov(\varepsilon_t, \varepsilon_{(t-1)-j})}_{0} + \rho \underbrace{Var(X_{t-1})}_{\frac{\sigma_\varepsilon^2}{1-\rho^2}} \\
&= \frac{\rho}{1-\rho^2}\sigma_\varepsilon^2.
\end{aligned}
$$

AR(1) Process: Autocorrelation function

- The **MA(∞)** representation of an **AR(1)** process (assuming $|\rho| < 1$):

$$X_t = \frac{\alpha}{1-\rho} + \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j},$$

  where $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$.

- **Autocovariance ($h \geq 2$):**

$$Cov(X_t, X_{t-h}) = \ldots = \frac{\rho^h}{1-\rho^2}\sigma_\varepsilon^2.$$

- **Autocovariance ($h \geq 1$):**

$$Cov(X_t, X_{t-h}) = \frac{\rho^h}{1-\rho^2}\sigma_\varepsilon^2.$$

AR(1) Process: Autocorrelation function

- The **MA(∞)** representation of an **AR(1)** process (assuming $|\rho| < 1$):

$$X_t = \frac{\alpha}{1-\rho} + \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j},$$

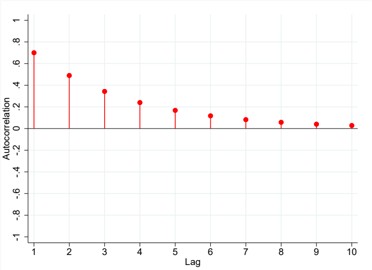
  where $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$.

- **Autocorrelation:**

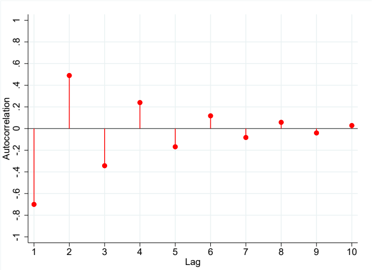
$$\begin{aligned} Corr(X_t, X_{t-h}) &= \frac{Cov(X_t, X_{t-h})}{\sqrt{Var(X_t)}\sqrt{Var(X_{t-h})}} \\ &= \frac{\frac{\rho^h}{1-\rho^2}\sigma_\varepsilon^2}{\sqrt{\frac{\sigma_\varepsilon^2}{1-\rho^2}}\sqrt{\frac{\sigma_\varepsilon^2}{1-\rho^2}}} \\ &= \rho^h. \end{aligned}$$

AR(1) Process: Autocorrelation function

- **ACF** for AR(1)



(a) $\rho_1 = 0.7$



(b) $\rho_1 = -0.7$

AR(1) Process: Summary

- Given $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$, an **AR(1)** process:

$$X_t = \alpha + \rho X_{t-1} + \varepsilon_t,$$

- An AR(1) process is **weakly stationary** if and only if $|\rho| < 1$.
  - characteristic function: $1 - \rho z^* = 0$ with $|z^*| > 1$
- An AR(1) process is not **strongly stationary**.
  - It is **strongly stationary** if $\varepsilon_s \sim IWN(0, \sigma_\varepsilon^2)$ for all $s \leq t$. (Why?)
- MA(∞) representation: $X_t = \frac{\alpha}{1-\rho} + \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j}$.
- Three moments:

  Mean: $E[X_t] = \frac{\alpha}{1-\rho}$

  Variance: $Var(X_t) = \frac{\sigma_\varepsilon^2}{1-\rho^2}$

  Covariance: $Cov(X_t, X_{t-h}) = \frac{\rho^h}{1-\rho^2}\sigma_\varepsilon^2$
  - If $\rho < 1$, $Cov(X_t, X_{t-h}) \to 0$ as $h \to \infty$.

Stationary Time-Series Processes

Auto Regressive Moving Average (ARMA) Process

ARMA(p,q) Process

Definition (Autoregressive Moving Average of Order (p,q))

A time series $\{X_t\}_{t=-\infty}^{\infty}$ is said to follow an ARMA(p,q) process if

$$X_t = \alpha + \rho_1 X_{t-1} + \rho_2 X_{t-2} + \ldots + \rho_p X_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-q},$$

where $\alpha, \rho_1, \ldots, \rho_p, \theta_1, \ldots, \theta_q \in \mathbb{R}$ are constants, and $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$.

- AR(p) component: **long-term dependence**
- MA(q) component: **short-term dependence**
- ARMA(p,q) simplifies to an AR(p) if $q = 0$, and an MA(q) if $p = 0$
- $\alpha, \rho_1, \ldots, \rho_p, \theta_1, \ldots, \theta_q, \varepsilon_s$: parameters

ARMA(p,q) Process

- Weak stationarity for ARMA(p,q) depends only on the AR(p) component:

$$X_t = \alpha + \underbrace{\rho_1 X_{t-1} + \rho_2 X_{t-2} + \ldots + \rho_p X_{t-p}}_{AR(p)} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-q}.$$

- The associated characteristic function:

$$1 - \rho_1 z - \rho_2 z^2 - \ldots - \rho_p z_p = 0.$$

- The ARMA(p,q) is weakly stationary.
  $\Longleftrightarrow$ All the roots of this equation are larger in magnitude than one.

ARMA(1,1) Process

- $\{X_t\}_{-\infty}^{\infty}$ is an **ARMA(1,1)** process if

$$X_t = \alpha + \rho X_{t-1} + \varepsilon_t + \theta\varepsilon_{t-1},$$

  where $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$.

- The **ARMA(1,1)** process is weakly stationary if and only if $|\rho| < 1$.
- The MA(∞) representation:

$$X_t = \frac{\alpha}{1-\rho} + \varepsilon_t + (\rho + \theta)\sum_{j=1}^{\infty} \rho^{j-1}\varepsilon_{t-j}$$

ARMA(1,1) Process

- The MA(∞) representation of an ARMA(1,1) Process:

$$
\begin{aligned}
X_t &= \alpha + \rho \underbrace{X_{t-1}}_{\alpha+\rho X_{t-2}+\varepsilon_{t-1}+\theta\varepsilon_{t-2}} + \varepsilon_t + \theta\varepsilon_{t-1} \\
&= \alpha + \rho(\alpha + \rho \underbrace{X_{t-2}}_{\alpha+\rho X_{t-3}+\varepsilon_{t-2}+\theta\varepsilon_{t-3}} + \varepsilon_{t-1} + \theta\varepsilon_{t-2}) + \varepsilon_t + \theta\varepsilon_{t-1} \\
&= \alpha + \rho\alpha + \rho^2(\alpha + \rho \underbrace{X_{t-3}}_{\alpha+\rho X_{t-4}+\varepsilon_{t-3}+\theta\varepsilon_{t-4}} + \varepsilon_{t-2} + \theta\varepsilon_{t-3}) + \varepsilon_t + (\rho+\theta)\varepsilon_{t-1} + \rho\theta\varepsilon_{t-2} \\
&\vdots \\
&= \alpha \underbrace{(1 + \rho + \rho^2 + \rho^3 + \dots)}_{\frac{\alpha}{1-\rho}} + \varepsilon_t + \underbrace{(\rho+\theta)\varepsilon_{t-1} + \rho(\rho+\theta)\varepsilon_{t-2} + \rho^2(\rho+\theta)\varepsilon_{t-3} + \dots}_{(\rho+\theta)\sum_{j=1}^{\infty}\rho^{j-1}\varepsilon_{t-j}} \\
&= \frac{\alpha}{1-\rho} + \varepsilon_t + (\rho+\theta)\sum_{j=1}^{\infty}\rho^{j-1}\varepsilon_{t-j}.
\end{aligned}
$$

ARMA(1,1) Process

- Given $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$, $\{X_t\}_{t=-\infty}^{\infty}$ is an **ARMA(1,1)** process if

$$X_t = \alpha + \rho X_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1},$$

- Using the MA(∞) representation, we can show

  Mean:

$$E[X_t] = \frac{\alpha}{1-\rho}$$

  Variance:

$$Var(X_t) = \left\{1 + \frac{(\rho+\theta)^2}{1-\rho^2}\right\}\sigma_\varepsilon^2$$

  Autocovariance:

$$Cov(X_t, X_{t-h}) = \begin{cases} \left\{(\rho+\theta) + \frac{(\rho+\theta)^2\rho}{1-\rho^2}\right\}\sigma_\varepsilon^2 & \text{for } h = 1 \\ \rho^{h-1} Cov(X_t, X_{t-1}) & \text{for } h \geq 2 \end{cases}$$

ARMA(1,1) Process

- Given $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$, $\{X_t\}_{t=-\infty}^{\infty}$ is an **ARMA(1,1)** process if

$$X_t = \alpha + \rho X_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1},$$

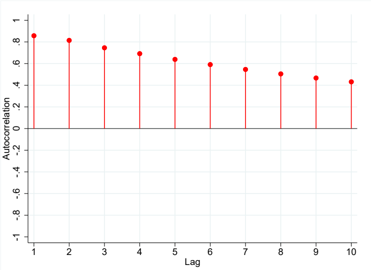
- Using the MA(∞) representation, we can show

  Autocorrelation:

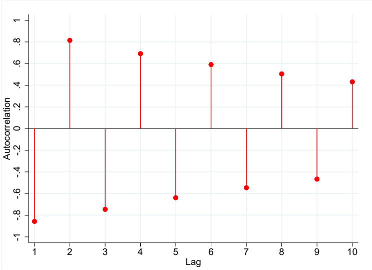
$$Corr(X_t, X_{t-h}) = \begin{cases} \frac{(\rho+\theta)(1+\rho\theta)}{1+2\rho\theta+\theta^2} & \text{for } h = 1 \\ \rho^{h-1} Corr(X_t, X_{t-1}) & \text{for } h \geq 2 \end{cases}$$

ARMA(1,1) Process

- **ACF** for ARMA(1,1)



(a) $\rho = 0.7, \theta = 0.7$



(b) $\rho = -0.7, \theta = -0.7$

Stationary Time-Series Processes

Gateway to Econometric Analysis

Recovering Parameters from Observables: Idea

- So far:
  - Knowledge: prameters (e.g., ρ, θ, σ_ε^2, etc)
  - Wants: moments of $\{X_t\}$ (e.g., mean, variance, autocovariance)
- In practice:
  - Knowledge: moments of $\{X_t\}$ (e.g., mean, variance, autocovariance)
  - Wants: prameters (e.g., ρ, θ, σ_ε^2, etc)
- Q. Can we recover parameters from moments?

Recovering Parameters from Observables: MA(q) Process

- Given $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$, an MA(1) process:

$$X_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}.$$

Mean: $E[X_t] = \mu$

Variacne: $Var(X_t) = (1 + \theta^2)\sigma_\varepsilon^2$

Autocovariance: $Cov(X_t, X_{t-h}) = \begin{cases} \theta\sigma_\varepsilon^2 & \text{for } h = 1 \\ 0 & \text{for } h \geq 2 \end{cases}$

- Q. Can we back out the parameters from the moments?

Recovering Parameters from Observables: MA(q) Process

- $\boldsymbol{\mu}$:
$$\mu = E[X_t]$$
$\hookrightarrow$ μ is recovered from $\{X_t\}$.

- $\boldsymbol{\theta}$:
$$\frac{\theta}{1+\theta^2} = \frac{Cov(X_t, X_{t-1})}{Var(X_t)}$$
$\hookrightarrow$ θ is recovered from $\{X_t\}$.

- $\boldsymbol{\sigma}_{\epsilon}^2$:
$$\sigma_\varepsilon^2 = \frac{Var(X_t)}{1+\theta^2} = \frac{Cov(X_t, X_{t-1})}{\theta}$$
$\hookrightarrow$ With θ recovered, σ_ε^2 is recovered from $\{X_t\}$.

Recovering Parameters from Observables: AR(p) Process

- Given $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$, an **AR(1)** process (with $|\rho| < 1$):

$$X_t = \alpha + \rho X_{t-1} + \varepsilon_t$$

Mean: $E[X_t] = \frac{\alpha}{1-\rho}$

Variance: $Var(X_t) = \frac{\sigma_\varepsilon^2}{1-\rho^2}$

Covariance: $Cov(X_t, X_{t-h}) = \frac{\rho^h}{1-\rho^2}\sigma_\varepsilon^2$

- Q. Can we back out the parameters from the moments?

Recovering Parameters from Observables: AR(p) Process

- $\boldsymbol{\rho}$:

$$\rho = \frac{Cov(X_t, X_{t-1})}{Var(X_t)}$$

$\hookrightarrow$ ρ is recovered from $\{X_t\}$.

- $\boldsymbol{\alpha}$:

$$\alpha = (1-\rho)E[X_t]$$

$\hookrightarrow$ With ρ recovered, α is recovered from $\{X_t\}$.

- $\boldsymbol{\sigma}_{\varepsilon}^2$:

$$\sigma_{\varepsilon}^2 = (1-\rho^2)Var(X_t) = \frac{1-\rho^2}{\rho}Cov(X_t, X_{t-1}) = E[\{X_t - (\alpha + \rho X_{t-1})\}^2]$$

$\hookrightarrow$ With α and ρ recovered, σ_{ε}^2 is recovered from $\{X_t\}$.

Recovering Parameters from Observables: ARMA(p, q) Process

- Given $\varepsilon_s \sim WN(0, \sigma_\varepsilon^2)$ for all $s \leq t$, an **ARMA(1,1)** process:

$$X_t = \alpha + \rho X_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1},$$

Mean: $E[X_t] = \frac{\alpha}{1-\rho}$

Variance: $Var(X_t) = \left\{1 + \frac{(\rho+\theta)^2}{1-\rho^2}\right\}\sigma_\varepsilon^2$

Autocovariance: $Cov(X_t, X_{t-h}) = \begin{cases} \left\{(\rho+\theta) + \frac{(\rho+\theta)^2\rho}{1-\rho^2}\right\}\sigma_\varepsilon^2 & \text{for } h = 1 \\ \rho^{h-1} Cov(X_t, X_{t-1}) & \text{for } h \geq 2 \end{cases}$

- Q. Can we back out the parameters from the moments?

Recovering Parameters from Observables: ARMA(p, q) Process

- $\boldsymbol{\rho}$:

$$\rho = \frac{Cov(X_t, X_{t-2})}{Cov(X_t, X_{t-1})}$$

$\hookrightarrow$ ρ is recovered from $\{X_t\}$.

- $\boldsymbol{\alpha}$:

$$\alpha = (1 - \rho)E[X_t]$$

$\hookrightarrow$ With ρ recovered, α is recovered from $\{X_t\}$.

- $\boldsymbol{\theta}$:

$$\frac{(\rho + \theta)(1 + \rho\theta)}{1 + 2\rho\theta + \theta^2} = \frac{Cov(X_t, X_{t-1})}{Var(X_t)}$$

$\hookrightarrow$ With ρ recovered, θ is recovered from $\{X_t\}$.

Recovering Parameters from Observables: ARMA(p, q) Process

- $\boldsymbol{\sigma}_{\varepsilon}^2$:

$$\sigma_{\varepsilon}^2 = \left\{1 + \frac{(\rho+\theta)^2}{1-\rho^2}\right\}^{-1} Var(X_t) = \left\{(\rho+\theta) + \frac{(\rho+\theta)^2\rho}{1-\rho^2}\right\}^{-1} Cov(X_t, X_{t-1})$$

$\hookrightarrow$ With ρ and θ recovered, σ_{ε}^2 is recovered from $\{X_t\}$.

Summary and Preview

Summary

- We explored more practical time-series processes: MA(q), AR(p), ARMA(p,q)
  - characteristic equation, MA(∞) representation
- Gateway to econometric analysis:
  - Can we "recover" the parameters from the observables?

Preview

- We will make the meaning of "recover" precise.
- We will start applying the linear regression framework to time-series data.
  - MA(q), AR(p), and ARMA(p,q) processes can be viewed as special cases.